BEFORE TUNING (BASELINE RANDOM FOREST MODEL)

RESULTS:

```
=== BASELINE RF ===
> cat("Balanced Accuracy:", round(baseline_confmatrix$byClass["Balanced Accurac
y"], 4),"\n")
Balanced Accuracy: 0.6848
> cat("Recall:", round(baseline_confmatrix$byClass["Sensitivity"], 4),"\n")
Recall: 0.9907
> cat("Specificity:", round(baseline_confmatrix$byClass["Specificity"], 4),"\n")
Specificity: 0.3788
> cat("F1:", round(baseline_confmatrix$byClass["F1"], 4),"\n")
F1: 0.9533
> cat("AUC-ROC:", round(auc_roc_baseline, 4),"\n")
AUC-ROC: 0.8027
```

AFTER TUNING:

(BEFORE THRESHOLD TUNING)

BEST TUNED PARAMETERS:

```
============ BEST TUNED RF =============
> cat("\nntree:",best_ntree,"\n")

ntree: 500
> cat("nodesize:",best_nodesize,"\n")
nodesize: 10
> cat("mtry:",best_mtry,"\n")
mtry: 2
> cat("ROC:", round(best_ROC,4),"\n")
ROC: 0.8323
```

RESULTS:

```
=== TUNED RF ===
> cat("Balanced Accuracy:", round(tuned_confmatrix$byClass["Balanced Accuracy"],
4),"\n")
Balanced Accuracy: 0.5224
> cat("Recall:", round(tuned_confmatrix$byClass["Sensitivity"], 4),"\n")
Recall: 0.9993
> cat("Specificity:", round(tuned_confmatrix$byClass["Specificity"], 4),"\n")
Specificity: 0.0455
> cat("F1:", round(tuned_confmatrix$byClass["F1"], 4),"\n")
F1: 0.9365
> cat("AUC-ROC:", round(best_ROC, 4),"\n")
AUC-ROC: 0.8323
```

AFTER THRESHOLD TUNING

RESULTS:

```
 THRESHOLD t = 0.2
> cat("Balanced Accuracy",cm_02$byClass["Balanced Accuracy"],"\n")
Balanced Accuracy 0.5
> cat("Recall:",cm_02$byClass["Sensitivity"],"\n")
Recall: 1
> cat("Specificity:", cm_02$byClass["Specificity"],"\n")
Specificity: 0
> cat("F1:",cm_02$byClass["F1"],"\n")
F1: 0.934044
> cat("\n THRESHOLD t = 0.5\n")

 THRESHOLD t = 0.5
> cat("Balanced Accuracy", cm_05$byClass["Balanced Accuracy"],"\n")
Balanced Accuracy 0.5223706
> cat("Recall:",cm_05$byClass["Sensitivity"],"\n")
Recall: 0.9992867
> cat("Specificity:",cm_05$byClass["Specificity"],"\n")
Specificity: 0.04545455
> cat("F1:", cm_05$byClass["F1"],"\n")
F1: 0.9364973
> cat("\n THRESHOLD t = 0.8\n")

 THRESHOLD t = 0.8
> cat("Balanced Accuracy", cm_08$byClass["Balanced Accuracy"],"\n")
Balanced Accuracy 0.7835884
> cat("Recall:",cm_08$byClass["Sensitivity"],"\n")
Recall: 0.9308131
> cat("Specificity:", cm_08$byClass["Specificity"],"\n")
Specificity: 0.6363636
> cat("F1:",cm_08$byClass["F1"],"\n")
F1: 0.9391868
```